Understanding Artificial Intelligence

Portfolio of Work (Summative Assessment)

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Component One: Predicting Customer Spending Behaviour in Streaming Services

Abstract

This project implemented supervised and unsupervised machine-learning techniques to predict the monthly spend and churn behaviour of 5,000 streaming-service customers. For preprocessing steps, missing data were imputed, numerical features were standardised and categorical variables were one-hot encoded. In terms of monthly spend, single-variable regression showed that Satisfaction Score was the strongest predictor, while a multiple linear regression using all numeric features achieved the highest accuracy ($R^2 \approx 0.886$; RMSE ≈ 3.21). A Random Forest and Artificial Neural Network performed slightly worse. For the churn prediction, the Random Forest classifier delivered the best performance (accuracy ≈ 0.98; ROC-AUC ≈ 0.99), while k-means clustering identified two customer segments with modest differences in behaviour. The findings generally demonstrate that linear regression and tree-based models provide usable analytics for the streaming service, which can support customer retention strategies.

1. Introduction

Streaming services have transformed consumption of media with subscription on-demand access. For long-term profitability, it is critical to understand how much customers spend and why they churn. A recent study on music streaming platforms has highlighted that up to 30–35% of new subscribers abandon the service within six months (Cheng and He, 2025). Customer retention is therefore a mission-critical function. Usman-Hamza et al. (2022) and Cheng and He (2025) also asserted that with regard to churn, it costs several times more to replace than to retain a subscriber.

Machine-learning models are increasingly used to predict churn and monthly spend in order to personalise offers and allocate marketing resources efficiently. However, a contemporary research study has shown that simple rule-based systems cannot capture complex behavioural

patterns, with ensemble methods or graph-based models often providing superior performance (Imani et al., 2025).

Accordingly, the goal herein is to analyse a synthetic streaming dataset which has 5 000 customers and 12 variables (six numeric and three categorical) with the aim of evaluating supervised models for predicting monthly spend and churn. This project also explored unsupervised clustering for customer segmentation.

2. Data Preparation

The dataset consists of 5 000 customers with six numeric and three categorical predictors, together with a churn indicator and the target variable Monthly Spend. The preliminary exploration revealed a right-skewed spend distribution, as shown in Figure 1.

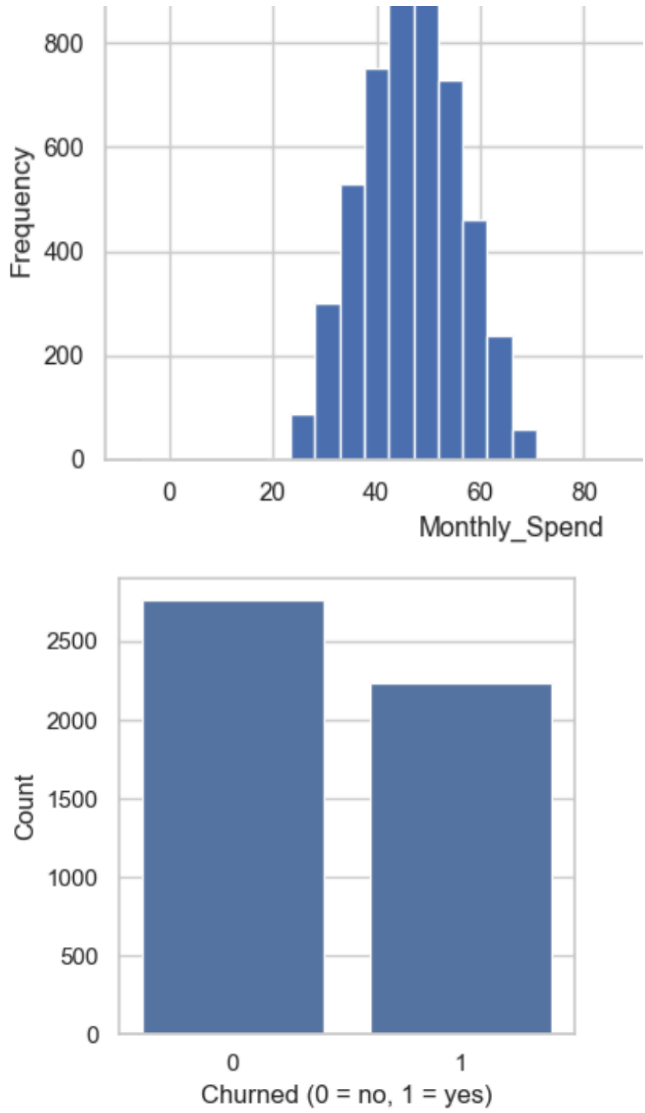


Figure 1: Distribution of Monthly Spend and churn rate

As part of the preprocessing steps, the missing values identified in the Age and Satisfaction Score were imputed using the median, numeric features were standardised while categorical variables were one-hot encoded. The dataset was then split into 4 000 training and 1 000 test cases for regression, with stratified splits used for classification. Imputation and scaling were thus critical for preprocessing because it has been shown that the poor handling of missing data or inconsistent scales can compromise classifier performance and clustering quality (Shadbahr et al., 2023; Wongoutong, 2024).

3. Q(a): Single-Variable Regression Models

Each of the numeric predictor was used individually to model the monthly spend using linear and polynomial (degree 2) regression. The results, which reports test R^2 and RMSE for each feature and indicates whether a linear or polynomial model is superior, are summarised in Figure 2.

	model_id	model_type	features	R2_test	RMSE_test
3	Single_Satisfaction_Score	Polynomial (degree=2)	Satisfaction_Score	0.534518	6.486895
1	Single_Subscription_Length	Linear	Subscription_Length	0.282882	8.051573
4	Single_Discount_Offered	Polynomial (degree=2)	Discount_Offered	0.044591	9.293517
2	Single_Support_Tickets_Raised	Linear	Support_Tickets_Raised	0.017760	9.423111
0	Single_Age	Polynomial (degree=2)	Age	-0.000793	9.511691
5	Single_Last_Activity	Polynomial (degree=2)	Last_Activity	-0.000924	9.512313

Figure 2: Test R^2 and RMSE for single-feature regression models (linear vs polynomial).

Satisfaction Score represents by far the strongest individual predictor: a degree-2 polynomial regression yields $R^2 \approx 0.53$ and RMSE≈6.49. Subscription Length comes as a distant second with $R^2 \approx 0.28$ and RMSE≈8.05. Other variables have little predictive power. The scatter plot in Figure 3 shows the non-linear relationship between Satisfaction Score and spending with a quadratic curve that fits the data better than a straight line.

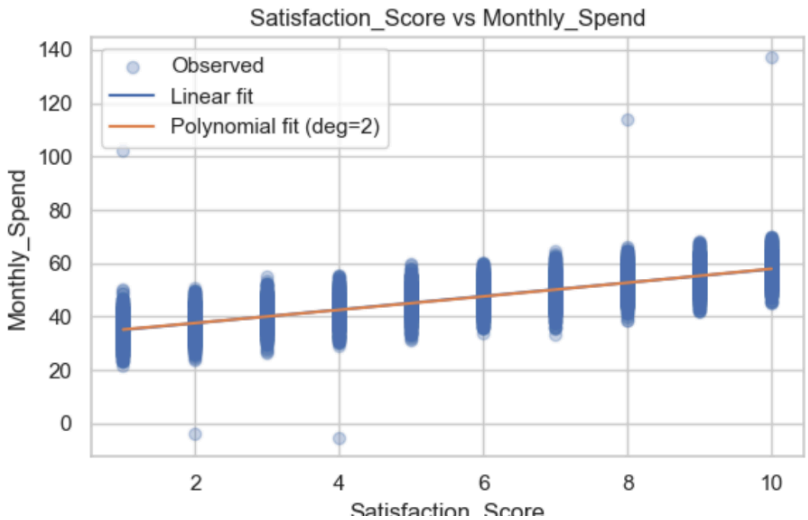


Figure 3: Scatter plots of Satisfaction Score vs Monthly Spend (top) and Discount Offered vs Monthly Spend (bottom). Curves show linear (blue) and quadratic (orange) fits.

The superiority of the polynomial model for the Satisfaction Score suggests that a non-linear relationship exists: i.e., customers with very high satisfaction tend to spend disproportionately more. Nevertheless, the single-feature models explain less than 54% of the variation and are insufficient for accurate prediction.

4. Q(b): Multi-Variable Regression Models

A multiple linear regression based on all the numeric features produced a substantial improvement: test R^2 increased to 0.886 and RMSE dropped to 3.21, and more than halving of the error of the best single-feature model. Train and test performance were similar, with little overfitting. Combining predictors helped the model to capture complementary information and yields a parsimonious but powerful model for spending.

5. Q(c): Mixed-Input Regression (Numeric + Categorical)

A Random Forest Regressor was trained on the one-hot-encoded dataset to incorporate both numeric and categorical predictors. The model achieved $R^2 \approx 0.867$ and RMSE≈3.47 on the test set, representing a slight reduction in accuracy when compared with the linear model. Feature

importance scores as shown in Figure 4 illustrate that Satisfaction Score and Subscription Length are dominant, while categorical predictors contribute very little.

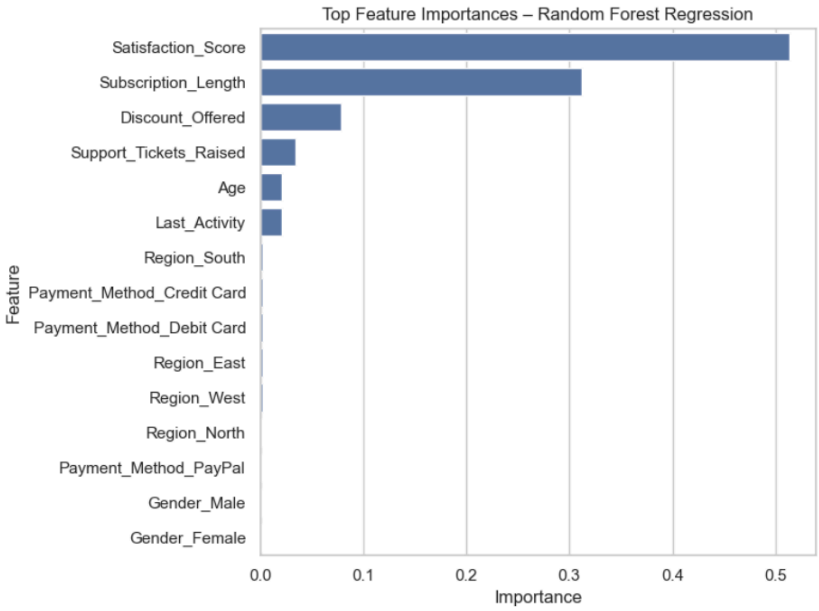


Figure 4: Random Forest feature importance scores for predicting monthly spend. Satisfaction Score and Subscription Length are by far the most influential predictors.

The inclusion of categorical variables does not significantly improve the prediction compared with the multiple linear regression model, which goes further to suggest that region, gender and payment method are not significantly related to spending in the dataset. It is important to note that decision forest models are generally widely advocated for customer analytics tasks because they handle complex interactions and class imbalance (Imani et al., 2025).

6. Q(d): Artificial Neural Network (ANN) Model

An Artificial Neural Network (ANN) with two hidden layers (64 and 32 neurons) was also trained to predict monthly spend. The model achieved test RMSE ≈ 3.38 and $R^2 \approx 0.874$, which is slightly worse than the performance of the linear model but comparable to the Random Forest.

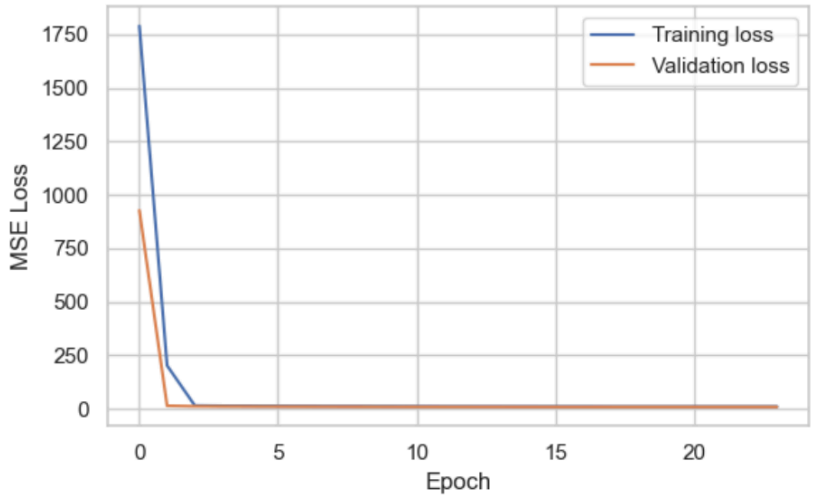


Figure 5: ANN training and validation loss across epochs (mean squared error). The steep decline in the first ten epochs reflects rapid learning, with early stopping preventing overfitting.

Comparative studies have shown that tree-based ensembles often outperform deep models on tabular data unless there are complex non-linear interactions (Fan and Waldmann, 2024). In the present situation, the ANN does not significantly outperform simpler models, which goes to suggest that the relationship between predictors and spend is largely linear.

7. Q(e): Best Model for Predicting Monthly Spend

Among the regression models, the multiple linear regression showed the highest accuracy ($R^2 \approx 0.886$, RMSE≈3.21). Random Forest and ANN models capture non-linearities but had slightly higher errors and signs of overfitting. The linear model therefore provides the best trade-off between accuracy, generalisation and interpretability for predicting spending.

8. Q(f): Classification Models – Churn Prediction

The churn indicator was predicted using three classifiers: Logistic Regression, Random Forest Classifier and Support Vector Machine with an RBF kernel. Table 1 summarises performance on the test set, and Figure 7 shows the ROC curve for the best model.

```
=== 8.2 Classification model comparison ===
```

	model_name	accuracy	precision	recall	f1_score	roc_auc
1	Random Forest Classifier	0.980	1.000000	0.955357	0.977169	0.989365
2	SVM (RBF)	0.941	0.920086	0.950893	0.935236	0.976716
0	Logistic Regression	0.801	0.762105	0.808036	0.784399	0.892894

Table 1: Classification performance metrics for churn prediction models (accuracy, precision, recall, F1-score and ROC-AUC).

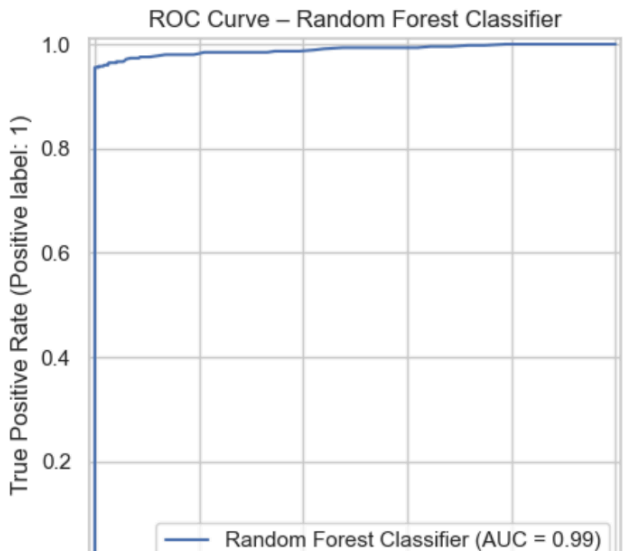


Figure 7: ROC curve for the Random Forest Classifier showing an AUC of approximately 0.99.

The Random Forest classifier achieved the highest accuracy (≈0.98) and F1-score (≈0.98), while the SVM and Logistic Regression models were less accurate. These results confirm that tree-based ensembles handle complex decision boundaries and class imbalance better than linear models (Imani et al., 2025). Recent research also recommends F1-score and the Matthews Correlation Coefficient for evaluating imbalanced classifications (Sujon et al., 2025). The Random Forest is therefore selected as the best classifier for predicting churn.

9. Q(g): k-Means Clustering – Customer Segmentation

Unsupervised clustering was performed on the scaled numeric features to identify customer segments. The k-means algorithm was evaluated for k between 2 and 10; the silhouette score

peaked at k = 2 (≈0.0933), so two clusters were selected. Figure 8 shows the cluster centroids and a PCA projection.

Cluster					
0	48.543300	5.631184	0.455307	44.960374	29.719872
1	44.687013	5.459750	0.440658	42.208962	29.688853

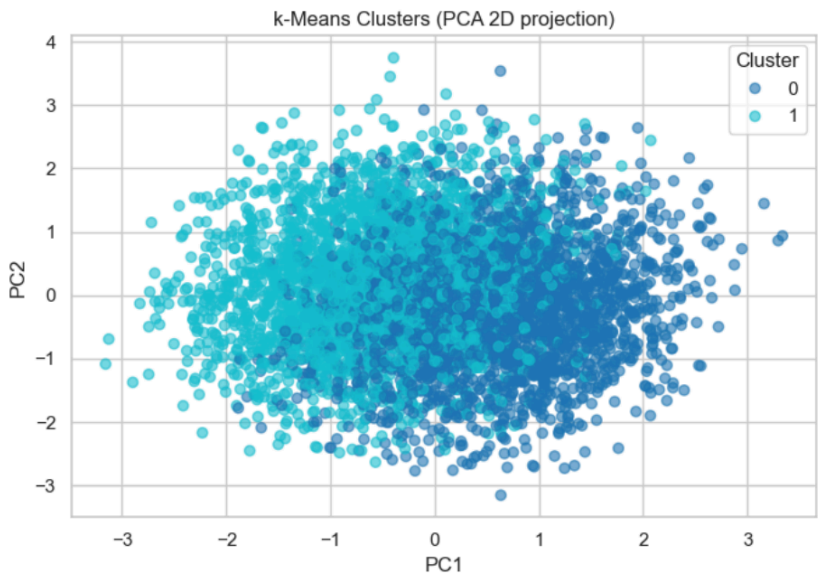


Figure 8: k-Means cluster profiles and two-dimensional PCA projection. Cluster 0 comprises 2 506 customers with higher average spend and satisfaction; Cluster 1 comprises 2 494 customers with slightly lower spend and satisfaction.

Cluster 0 customers have slightly higher spend, satisfaction and age than Cluster 1 customers. The differences are modest, reflecting the synthetic nature of the dataset. Scaling features prior to clustering improves cluster quality (Wongoutong, 2024).

10. Q(h): Alternative Clustering Algorithm

An agglomerative hierarchical clustering algorithm using Ward linkage was applied as an alternative.

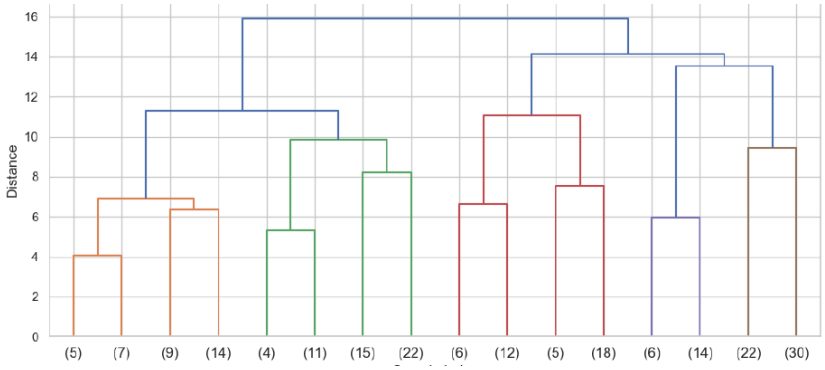


Figure 9: Dendrogram from hierarchical clustering. The colour bands highlight two main clusters (heights indicate linkage distances).

Hierarchical clustering produced two clusters of sizes 2 293 and 2 707 with a silhouette score of 0.0579—lower than k-means. The clusters were less balanced and overlapped more, reducing interpretability. Overall, k-means provides more meaningful segmentation for this dataset.

11. Conclusion

The analysis herein compared regression, classification and clustering models on a synthetic streaming dataset. The result showed that the multiple linear regression model delivered the highest accuracy, surpassing single-feature models and even more complex Random Forest and ANN models. Satisfaction Score and Subscription Length were the most influential predictors. The Random Forest classifier achieved the best churn prediction, highlighting the value of tree-based ensembles for imbalanced classification. k-means clustering identified two customer segments with modest differences; hierarchical clustering performed worse. Future work could incorporate additional behavioural features, explore profit-oriented evaluation metrics and investigate graph-based models that integrate relational information.

Component Two: Vehicle Damage Insurance Claim Verification Using CNN

Abstract

This study investigates the use of a convolutional neural network (CNN) to automate verification of vehicle-damage insurance claims across six categories. A baseline CNN performed poorly, but applying regularisation techniques, data augmentation and hyper-parameter tuning improved validation accuracy to approximately 0.48. The final model showed no evidence of overfitting but remained capacity-limited due to dataset imbalance. The findings highlight both the potential and constraints of CNN-based claim verification and recommend transfer learning and larger datasets for real-world deployment.

1. Introduction

Motor insurers traditionally rely on manual inspection to verify claims of vehicle damage, a process that demands physical examinations and evaluation of numerous photographs by trained appraisers. This practice is costly and slow, as illustrated by Pérez-Zarate et al. (2024). Automating claim verification promises substantial savings and faster settlements.

A typical Deep earning technique like CNN applies successive convolutional layers to learn low-level features such as edges and textures, and then deeper layers capture complex shapes; pooling layers reduce dimensionality while preserving salient information, and the resulting features are fed into fully-connected layers for classification (Maurício, Domingues and Bernardino, 2023). CNNs therefore form a natural choice for classifying damage photographs sent by claimants.

The objective of this report is to design and analyse a CNN for automatic verification of vehicle-damage claims, assess how well the network distinguishes between different damage categories (crack, scratch, tyre flat, dent, glass shatter and lamp broken) and evaluate the effectiveness of regularisation and hyper-parameter tuning.

2. Dataset & Pre-Processing

2.1 Dataset description

The dataset used in this exercise originates from the Kaggle competition Vehicle Damage Insurance Verification and contains six classes: crack, scratch, tyre flat, dent, glass shatter and lamp broken. The training set was stratified into 5760 images (80 %) and the validation set into 1440 images (20 %); no separate test set is indicated, so evaluation was performed on the validation data. Figure 1 shows both the numeric distribution and the bar chart of the six classes.

	label	damage_type	count	percentage
0	1	crack	171	2.375000
1	2	scratch	2349	32.625000
2	3	tire flat	534	7.416667
3	4	dent	2079	28.875000
4	5	glass shatter	1185	16.458333
5	6	lamp broken	882	12.250000

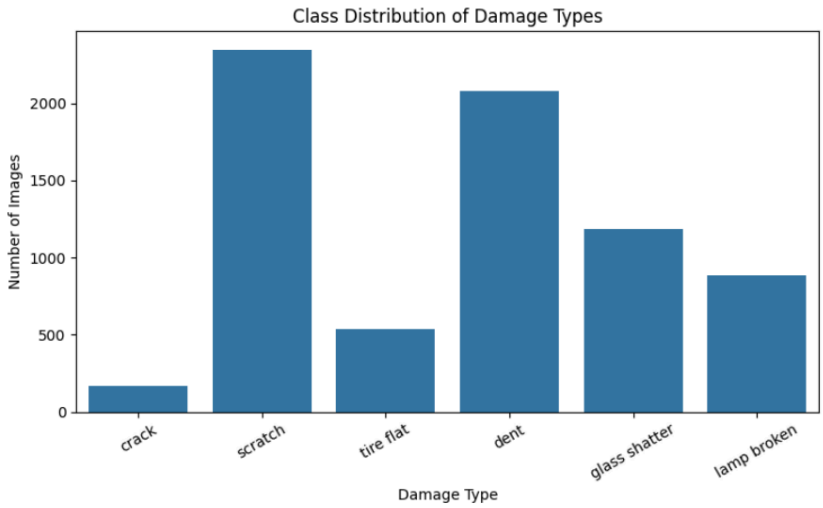


Figure 1: Distribution of damage categories in the training dataset.

2.2 Pre-processing and augmentation

All images were resized to 224×224 pixels and rescaled by dividing pixel values by 255 to normalise them into the $[0, 1]$ range. Normalisation accelerates convergence and stabilises gradient updates, as noted in the literature. For training, an `ImageDataGenerator` applied data augmentation comprising random rotations of 15°, horizontal and vertical translations of up to 10 % of the image size, zooming by 10 %, and horizontal flipping. A recent study has

demonstrated that such augmentations improve generalization (Omoniyi et al., 2025). Figure 2 illustrates some sample images for each class after pre-processing.

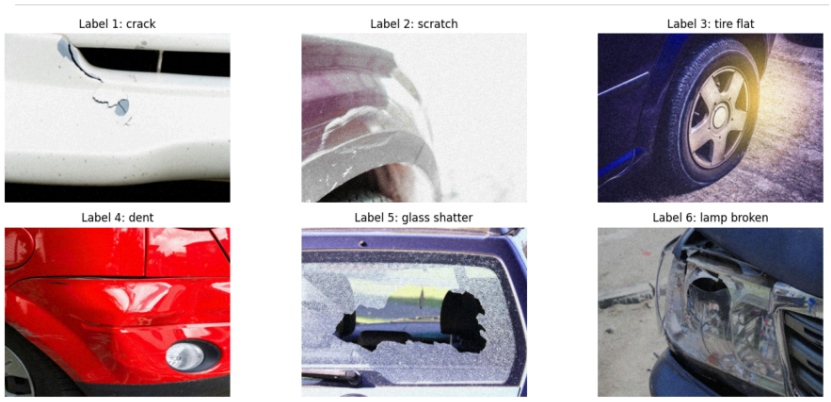


Figure 2: Example training images after resizing and augmentation

3. Q(a): CNN Model Architecture Description & Justification

3.1 Baseline architecture

The initial CNN consisted of three convolutional blocks. The first block used 32 filters with 3 × 3 kernels, followed by batch normalisation and a 2 × 2 max-pooling layer. The second and third blocks doubled the number of filters to 64 and 128, respectively, with identical kernel sizes, batch normalisation and pooling operations. After flattening, a dense layer with 256 units and batch normalisation was applied, followed by a softmax layer producing probabilities for the six classes. The model contained approximately 25.8 million parameters. The network was trained using the Adam optimiser with a learning rate of 10^{-3} and a categorical cross-entropy loss function. Figure 3 reproduces the architecture summary from the analysis output.

```
=== 4.1 Baseline CNN Summary ===
Model: "baseline_cnn"
```

Layer (type)	Output Shape
input_layer (InputLayer)	(None, 224, 224, 3)
conv2d (Conv2D)	(None, 224, 224, 32)
batch_normalization (BatchNormalization)	(None, 224, 224, 32)
max_pooling2d (MaxPooling2D)	(None, 112, 112, 32)
conv2d_1 (Conv2D)	(None, 112, 112, 64)
batch_normalization_1 (BatchNormalization)	(None, 112, 112, 64)
max_pooling2d_1 (MaxPooling2D)	(None, 56, 56, 64)
conv2d_2 (Conv2D)	(None, 56, 56, 128)
batch_normalization_2 (BatchNormalization)	(None, 56, 56, 128)
max_pooling2d_2 (MaxPooling2D)	(None, 28, 28, 128)
flatten (Flatten)	(None, 100352)
dense (Dense)	(None, 256)
batch_normalization_3 (BatchNormalization)	(None, 256)
dense_1 (Dense)	(None, 6)

```
Total params: 25,787,078 (98.37 MB)
Trainable params: 25,786,118 (98.37 MB)
Non-trainable params: 960 (3.75 KB)
```

Figure 3: Baseline CNN architecture summary.

3.2 Regularised architecture

To improve generalisation, a regularised version of the CNN was developed. This architecture preserved the three convolutional blocks but added dropout layers after each block and before the final dense layer. Dropout randomly deactivates neurons during training and is widely employed to prevent co-adaptation of weights; overfitting remains a major difficulty in deep networks, and dropout is a simple and effective regulariser (Çapkan and Yeşildirek, 2025). Weight decay (L2 regularisation) was also applied to convolutional and dense layers to penalise large weights, and batch normalisation was retained. The model was trained using early stopping with patience to halt training when the validation loss failed to improve. This strategy

has been shown to maintain generalisation across diverse datasets (Hayat et al., 2025). Figure 4 shows the architecture summary of the regularised model.

Layer (type)	Output Shape
input_layer_1 (InputLayer)	(None, 224, 224, 3)
conv2d_3 (Conv2D)	(None, 224, 224, 32)
batch_normalization_4 (BatchNormalization)	(None, 224, 224, 32)
max_pooling2d_3 (MaxPooling2D)	(None, 112, 112, 32)
dropout (Dropout)	(None, 112, 112, 32)
conv2d_4 (Conv2D)	(None, 112, 112, 64)
batch_normalization_5 (BatchNormalization)	(None, 112, 112, 64)
max_pooling2d_4 (MaxPooling2D)	(None, 56, 56, 64)
dropout_1 (Dropout)	(None, 56, 56, 64)
conv2d_5 (Conv2D)	(None, 56, 56, 128)
batch_normalization_6 (BatchNormalization)	(None, 56, 56, 128)
max_pooling2d_5 (MaxPooling2D)	(None, 28, 28, 128)
dropout_2 (Dropout)	(None, 28, 28, 128)
flatten_1 (Flatten)	(None, 100352)
dense_2 (Dense)	(None, 256)
batch_normalization_7 (BatchNormalization)	(None, 256)
dropout_3 (Dropout)	(None, 256)
dense_3 (Dense)	(None, 6)

Total params: 25,787,078 (98.37 MB)

Trainable params: 25,786,118 (98.37 MB)

Non-trainable params: 960 (3.75 KB)

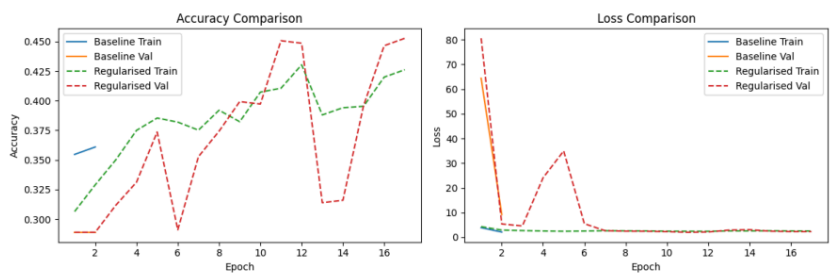
Figure 4: Regularised CNN architecture incorporating dropout and weight decay.

4. Q(b): Regularisation Methods and Their Impact

Several regularisation methods were applied and their impacts are evident in the learning curves. First, data augmentation (discussed in Section 2) expanded the training set and introduced geometric variability. Second, dropout layers were inserted after each convolutional

block and before the dense layer. Third, L2 weight decay was applied to convolutional and dense layers. Fourth, batch normalisation was used throughout the network. Finally, early stopping monitored the validation loss and stopped training when it failed to improve for several epochs.

Figure 5 compares the training and validation accuracy and loss for the baseline and regularised models. The baseline network achieved a validation accuracy of only 0.289 and a high validation loss of 9.20 after two epochs, indicating severe underfitting. In contrast, the regularised model, trained for 20 epochs with early stopping, reached a validation accuracy of approximately 0.451 and a validation loss of 2.00. The accuracy curves show that the regularised model learns more effectively and narrows the gap between training and validation performance, while the loss curves demonstrate reduced overfitting. Dropout and weight decay thus had a substantial positive impact on generalisation, aligning with literature reports that dropout improves robustness in deep networks (Çapkan and Yeşildirek, 2025).




```
Regularised validation accuracy: 0.4507
Regularised validation loss: 1.9999

Validation performance comparison:
```

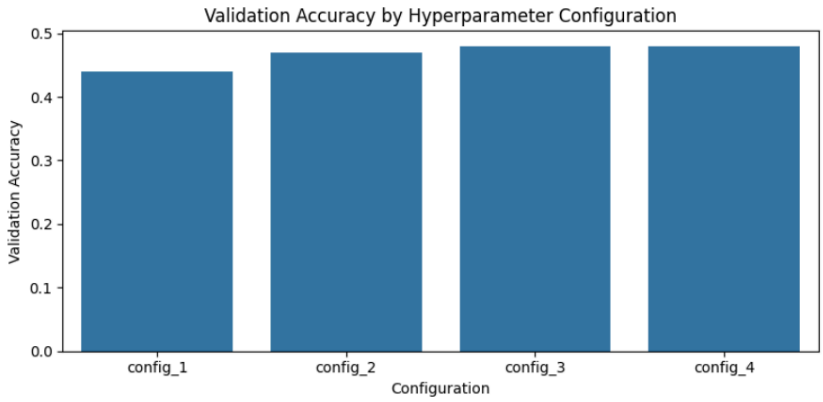
	model	val_accuracy	val_loss
0	baseline	0.288889	9.202246
1	regularised	0.450694	1.999926

Figure 5: Comparison of baseline and regularised training/validation curves.

5. Q(c): Hyper-parameter Tuning & Performance Impact

Hyper-parameters such as learning rate and number of filters determine how effectively a network learns. Four configurations were evaluated by varying the initial learning rate (10^{-3} vs 5×10^{-4}) and the base number of filters (32 vs 64). Each model was trained for 12 epochs with early stopping. As shown in Figure 6, the configuration with a learning rate of 10^{-3} and 64 filters

(config_3) achieved the highest validation accuracy of 0.4799 and a validation loss of 2.07. Lower learning rates slowed convergence and resulted in slightly inferior performance. Increasing the number of filters from 32 to 64 improved feature representation capacity and produced higher accuracy.

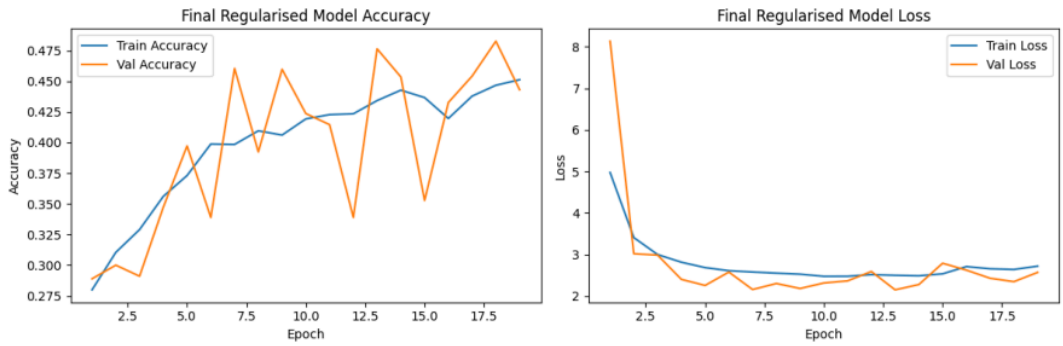



```
Best configuration:
{'name': 'config_3', 'learning_rate': 0.001, 'batch_size': 32, 'base_filters': 6
4, 'val_accuracy': 0.4798611104488373, 'val_loss': 2.0718958377838135}
```

Figure 6: Validation accuracy across hyper-parameter configurations.

6. Q(d): Overfitting Assessment

The final model, trained using the best hyper-parameters (learning rate 10^{-3}, base filters 64) and regularisation, was run for up to 25 epochs with early stopping. Figure 7 plots training and validation accuracy and loss. The training accuracy plateaued around 0.451 with a training loss of 2.72, while the best validation accuracy reached 0.476 with a validation loss of 2.15. Notably, the validation accuracy slightly exceeded the training accuracy, implying that the regularisation techniques effectively constrained the model and prevented overfitting. No divergence between the curves is observed; instead, both stabilise at similar values. Therefore, the final model does not exhibit classic overfitting but rather seems under-powered to fully fit the training data, likely due to the limited dataset size and class imbalance. Early stopping with a patience of six epochs halted training before the network began to memorise noise. The combination of dropout, weight decay, and data augmentation mitigated overfitting and produced a model that generalises as well as possible on the available data.




```
Final training accuracy (last epoch): 0.4512
Final training loss (last epoch): 2.7233
Final validation accuracy (best): 0.4764
Final validation loss (best): 2.1531
```

Figure 7: Training and validation curves for the final regularised model.

7. Model Evaluation & Accuracy Reporting

7.1 Quantitative metrics

Evaluation on the validation set yielded an overall accuracy of 0.48 and a macro-averaged F1-score of 0.40, as reported in the classification report (Figure 8).

```
Found 1440 validated image filenames belonging to 6 classes.
45/45 ━━━━━━━━━━━━━━━━━━━━ 39s 861ms/step

Classification report (validation set):
               precision    recall  f1-score   support

        crack       0.00      0.00      0.00        34
         dent       0.38      0.94      0.54       416
glass shatter       0.75      0.75      0.75       237
  lamp broken       0.56      0.30      0.39       176
      scratch       1.00      0.01      0.01       470
    tire flat       0.92      0.57      0.71       107

     accuracy                           0.48      1440
    macro avg       0.60      0.43      0.40      1440
 weighted avg       0.70      0.48      0.38      1440
```

Figure 8: Classification report showing precision, recall and F1-scores for each damage class.

7.2 Confusion matrix and error analysis

The confusion matrix (Figure 9) reveals the patterns of misclassification. Many crack and scratch images were misidentified as dents, reflecting the dominance of the dent class. The model correctly identified most glass-shatter and lamp-broken instances, indicating that these classes have distinctive features. However, the confusion between scratches and dents underscores the challenge of distinguishing superficial surface damage from deeper dents, particularly when the

dataset is imbalanced, and the network has limited capacity. Misclassifications of tyre flat images were also frequent, suggesting that the network may not adequately capture features of deflated tyres.

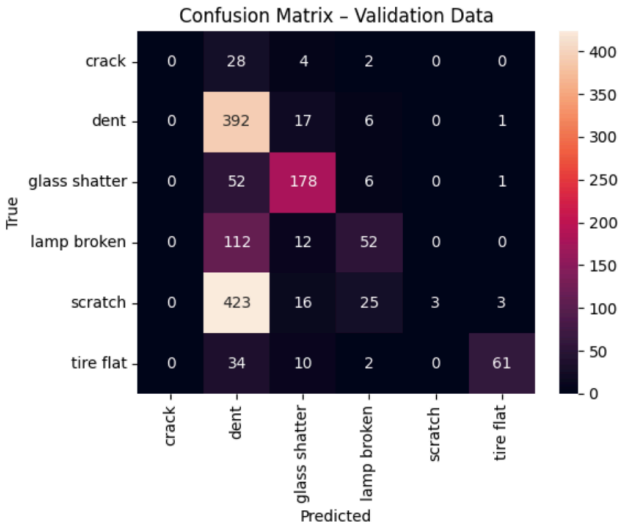


Figure 9: Confusion matrix of the final CNN model

8. Conclusion

This study constructed and evaluated a convolutional neural network to classify vehicle damage images for insurance claim verification. Using a stratified dataset of six damage categories, the baseline CNN achieved poor performance, but regularisation with dropout, weight decay and early stopping substantially improved validation accuracy. Nevertheless, the final model exhibits limited accuracy due to dataset imbalance and constrained model capacity. Future work should explore transfer learning with pre-trained networks, more comprehensive data collection, and advanced augmentation techniques to achieve the robustness required for real-world insurance applications.

Component Three: Fair AI in Healthcare

1. Introduction

Artificial intelligence (AI) is rapidly transforming the delivery of healthcare in the areas of diagnostic imaging, risk prediction and triage decisions. However, there is evidence to suggest that the data that is generally used to train these systems often encodes historical inequities (Grote and Keeling, 2022; Chen et al., 2023; Cross, Choma and Onofrey, 2024). For example, the insufficient sample sizes for certain patient groups usually result in poor algorithmic performance and clinically unmeaningful predictions (Cross, Choma and Onofrey, 2024). To address these ethical risks, this report presents an analysis on the theme of fair AI in healthcare, especially in terms of technical challenges, ethical frameworks, and potential innovations.

2. Technical challenges of fair AI in healthcare (Q a)

Data imbalance and under-representation: Clinical datasets most often contain far fewer samples from minority ethnic groups or patients with rare conditions. Chen et al. (2023) noted in this regard that the distribution of outcomes across ethnicities can be skewed, like in the case of underserved Hispanic and Black patients that experience delayed referrals because of health datasets that have internalised social biases (Chen et al., 2023). Cross, Choma and Onofrey (2024) has similarly shown that insufficient sample sizes lead to algorithm underestimation and clinically unmeaningful predictions (Cross, Choma and Onofrey, 2024). Such imbalances can cause models to generalise poorly and perpetuate unequal care for under-represented groups.

Label and historical bias: Labels that are used to train AI may sometimes reflect implicit cognitive biases or substandard care practices (Cross, Choma and Onofrey, 2024). For example, the pain scores recorded by clinicians may underestimate the pain in Black patients because of racial stereotypes (Grote and Keeling, 2022). When biased labels are used as ground truth in this way, even sophisticated models may propagate injustice.

Domain and dataset shift: Furthermore, models trained in one hospital may perform poorly in another because of factors that are related to demographic, genetic or procedural differences. Such dataset shift can come from variations in population demographics, image-acquisition techniques and disease prevalence (Chen et al., 2023). Even subgroup shifts are capable of causing black box models to leak protected attributes and misdiagnose patients when deployed across regions (Chen et al., 2023).

Model opacity and explainability: Generally speaking, deep-learning models often operate as opaque "black boxes" that clinicians cannot question. Without further explanations, it is difficult to detect hidden biases or to understand why an AI recommends a certain course of action. Although post-hoc explainability techniques using techniques like SHAP and LIME can provide local explanations, these may not align with clinical reasoning. Therefore, inherently interpretable models need to be prioritised for high-stakes decisions (Johannssen and Chukhrova, 2025). It is important to note that interpretable models sometimes underperform on complex data. Consequently, achieving both high performance and transparency presents a technical challenge that needs to be tackled.

3. Ethical frameworks and models: strengths and limitations (Q b)

A number of ethical frameworks have been proposed in relation to fair AI in healthcare. Generally, principles-based frameworks emphasise fairness, accountability, transparency and explainability. Although these frameworks generally raise awareness of bias and encourage documentation, they can often lack the operational guidance need to measure and achieve fairness in practice.

Affirmative algorithms and collaboration models: Grote & Keeling (2025) argued in this regard that the fairness of algorithmic outputs should not be the sole locus of moral evaluation. The authors have suggested that algorithms that are trained on diverse datasets might permissibly perform better for disadvantaged groups, provided that the final clinician-algorithm decision is fair. This affirmative approach highlights the role of human–AI collaboration which recognises that clinicians may override and adjust the recommendations of algorithms. The strength of this

framework lies in focusing on patient outcomes rather than abstract metrics, even though it poorly assumes that clinicians will have the time and expertise to interpret algorithmic advice.

Risk-based regulatory frameworks: Emerging standards such as the Fairness of Artificial Intelligence Recommendations in healthcare (FAIR) statement emphasise algorithm audits, transparency reports and post-market surveillance (Ueda et al., 2023). These frameworks integrate ethical principles into the regulatory processes, which can help developers to demonstrate compliance. However, it is noteworthy to indicate that audits can be costly, and regulatory guidance may lag behind technical innovation. Furthermore, reliance on self-reported audits may fail to detect subtler forms of bias.

4. Innovation and technical solutions (Q c)

To address the limitations identified above, the following is a suite of technical innovations that can be integrated into a governance framework.

1. **Robust bias-mitigation pipelines and continuous monitoring and adaptive re-training (Chen et al., 2023):** Bias needs to be addressed at all stages of the modelling pipeline. With pre-processing methods such as importance reweighting and resampling, sample-selection bias can be corrected bias by increasing the weight of under-represented groups. In-processing techniques, like adversarial training that penalises the learning of protected-attribute signals, can also promote invariant representations. Post-processing methods can also adjust the decision thresholds to achieve equalised odds, though careful calibration is required to avoid harming certain groups. Combining these approaches in a pipeline can aid developers to test and compare fairness–accuracy trade-offs and to tailor mitigation strategies to specific clinical tasks. Furthermore, techniques such as group distributionally robust optimisation (GroupDRO) can also address subpopulation shifts, but continuous re-training with new data from diverse populations is essential.

2. **Human-in-the-loop decision support (Grote and Keeling, 2022):** Clinicians need to remain central in decision-making. Interfaces need to also present algorithmic predictions alongside confidence intervals and explanations, which should allow

clinicians to contextualise recommendations with patient histories. System designs should invariably support contestability such that clinicians can override algorithmic outputs and provide feedback for model updates. This all are in alignment with the affirmative collaboration model that mitigates overreliance on imperfect algorithms.

3. **Explainable AI (XAI) for fairness and trust (Johannssen and Chukhrova, 2025):** XAI techniques can enhance fairness by making visible hidden biases. Johannssen and Chukhrova (2025) noted in this regard that XAI ensures that decisions are not only accurate but also fair, reliable, unbiased and reasonable. Explainability can help all stakeholders to question how and why models arrive at specific decisions, which can help in the detection of hidden biases. Furthermore, integrating XAI with fairness audits can also show discriminatory patterns and support accountability. Healthcare organisations generally need to adopt a combination of interpretable models for high-stakes decisions and post-hoc explanation methods (SHAP, LIME, counterfactuals) when complex models are necessary.

5. Conclusion

Fairness in healthcare AI is not a passive endeavour but a continual effort that involves identifying and mitigating bias. To move towards trustworthy and fair AI, healthcare organisations need to implement robust bias-mitigation pipelines, monitor models for fairness drift, support human-in-the-loop decision-making, adopt XAI techniques and equitably govern data.

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